Marcelo Cambraia Villela

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Full Stack **.NET** developer with over 2 years of experience on Participa DF, a mission-critical government platform in Brasília with over one million registered users. I work with **C#**, **ASP.NET** (MVC and Web API), **SQL Server** and **Angular**, in a codebase built on DDD and CQRS/MediatR. I designed an event-driven integration with **RabbitMQ**, implemented Gov.br sign-in with **OAuth 2.0 / OpenID Connect**, and troubleshoot production issues down to the root cause. On my own project, I shipped **PIX** payments to production. Associate Degree in Systems Analysis and Development; advanced English, remote-ready.

Technical Skills

- **Back End**: C#, .NET, ASP.NET Core, ASP.NET MVC, ASP.NET Web API, REST APIs, CQRS/MediatR, DDD, LINQ, Minimal APIs; SOLID, Clean Code and design patterns.
- **Data**: Microsoft SQL Server / T-SQL (daily use), Entity Framework Core, NHibernate and Dapper; MongoDB and PostgreSQL; query optimization, transactions and concurrency.
- **Integration and Resilience**: RabbitMQ (publish/subscribe, dead-letter and retry), Transactional Outbox pattern, Polly (retry, circuit breaker, timeout, fallback), distributed caching.
- **Security and Identity**: OAuth 2.0, OpenID Connect, PKCE, JWT, OWASP Top 10, mTLS, Row Level Security and LGPD (Brazil's data protection law).
- **Front End**: Angular, NgRx, AngularJS, React, Next.js, TypeScript, JavaScript, HTML and CSS.
- **Quality and Tools**: xUnit, Cypress, Swagger/OpenAPI, Postman, code review; Git (GitLab and GitHub), CI/CD with Jenkins, Docker, Google Cloud Run; Scrum and Kanban; Claude Code, GitHub Copilot and Cursor in daily development.

Professional Experience

Office of the Comptroller General of the Federal District (CGDF) **Brasília, Brazil**
Full Stack Developer *Nov 2024 - Present*
Intern from Nov 2024 to Feb 2025; hired full-time in Mar 2025.

- Builds and maintains **Participa DF**, a mission-critical government platform with over one million registered users, using **C# .NET** (ASP.NET MVC and Web API) with CQRS/MediatR, NHibernate, **SQL Server** and front ends in AngularJS and Angular.
- Works in a pragmatic **DDD** architecture: bounded contexts per service, rich entities, value objects, domain events via MediatR and the notification pattern.
- Designed and implemented from scratch an **event-driven asynchronous integration via RabbitMQ** between two government systems, with dedicated queues, payload contracts and a dead-letter/retry strategy; evaluated the **Transactional Outbox** pattern.
- Fixed a subtle **idempotency** issue caused by NHibernate auto-flush, and implemented transactional locking to restrict operations on an entity while awaiting a response from an external system.
- Identified through **decompilation** that a third-party library was silently swallowing BackgroundService exceptions, masking real connectivity failures.
- Implemented the integration with **Gov.br** (Brazil's federal single sign-on) using OAuth 2.0 / OpenID Connect, Authorization Code Flow and PKCE.
- Maintained **People Check**, an identity-verification service on **MongoDB** used in citizen authentication, and its .NET integration with Polly resilience policies, API-key authentication and a distributed cache.
- Maintained the unified search API of the **DF Transparency Portal** over MongoDB collections, with text search, direct lookup by individual and company tax IDs, and pagination.
- Migrated features from **AngularJS to Angular**, preserving legacy behavior, with state management in **NgRx**.

- Optimized SQL Server queries with **Dapper**, fixed production bugs and takes part in code reviews through GitLab merge requests.

Projects

- **github.com/MarceloCambraia/apabb-app**
  APABB App, donation and volunteering platform
  - Payment integration with Banco do Brasil APIs: **PIX in production** via an mTLS proxy on Google Cloud Run, bank slips (boleto) and BB Pay checkout.
  - React/TypeScript front end, serverless back end on Supabase (PostgreSQL), multi-role Row Level Security, E2E tests with Cypress and LGPD compliance.
- **Personal project**
  MetaAssist, AI-powered systematic review and meta-analysis tool
  - Backend API in **ASP.NET Core 8** (Minimal APIs) with Supabase/PostgreSQL storage and a React/Vite front end.
  - Integration with LLMs via OpenRouter for automated screening of scientific studies.

Education

- **UniCEUB** **Brasília, Brazil**
  Associate Degree (Tecnólogo) in Systems Analysis and Development *2026*
  Completed in 2026.

Languages

- Portuguese (native), English (advanced).

References

- Available upon request.